

Gravitation

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Q1. A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy will be _____ (g is acceleration due to gravity at the surface of earth).

JEE Main 2026 (Online) 5th April Evening Shift

✓ **Answer: $2mgR_e/3$**

Solution:

PE at the surface = $-GMm/R_e$.

At height $h = 2R_e$, the distance from the centre is $3R_e$, so $PE_{\text{final}} = -GMm/(3R_e)$.

$\Delta U = PE_{\text{final}} - PE_{\text{initial}} = -GMm/(3R_e) + GMm/R_e = 2GMm/(3R_e)$.

Since $g = GM/R_e^2$, $GM = gR_e^2$, so

$\Delta U = 2gR_e^2 m/(3R_e) = 2mgR_e/3$.

Q2. When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface, the change in g is approximately $\alpha\%$. The value of α is _____. (Take radius of the earth = 6400 km.)

JEE Main 2026 (Online) 5th April Morning Shift

✓ **Answer: 0.25**

Solution:

Below the surface (depth d): $g_{\text{below}} = g(1 - d/R) = g(1 - 16/6400) = 0.9975g$.

Above the surface (height $h \ll R$): $g_{\text{above}} \approx g(1 - 2h/R) = g(1 - 32/6400) = 0.995g$.

Change in $g = g_{\text{above}} - g_{\text{below}} = (0.995 - 0.9975)g = -0.0025g$, i.e. a decrease of 0.25%. So $\alpha = 0.25$.

Source: JEE Main 2026 (NTA official releases, physics section) · formulaverse.in